



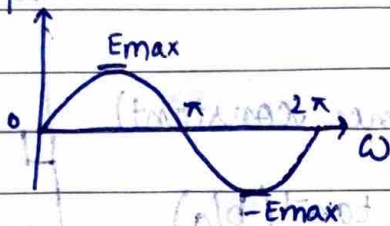
Unit → 2

→ Generation of single phase AC

Set of conductors move relatively in presence of magnetic field (uniform) to induce emf which produces alternating current.

$$e = - \frac{d\phi}{dt}$$

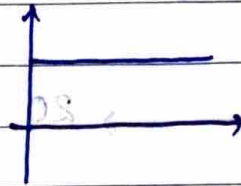
emf



AC

(Diff values at diff time)

DC



→ Representation of single phase AC

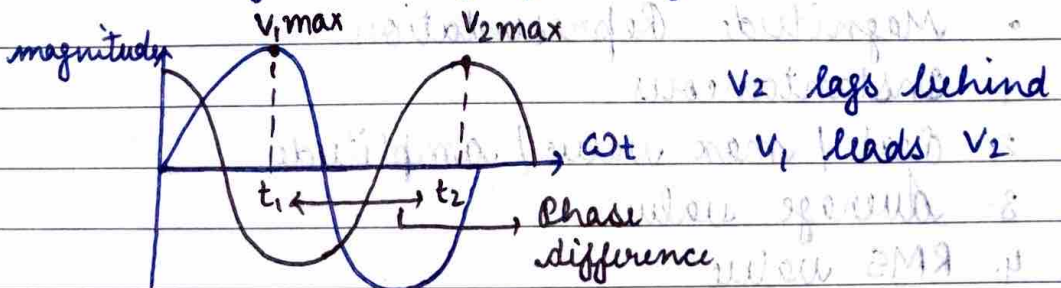
• Graphical form

1. wave form

Traced between two axes

x axis → phase angle / time duration of quantity

y axis → magnitude of quantity

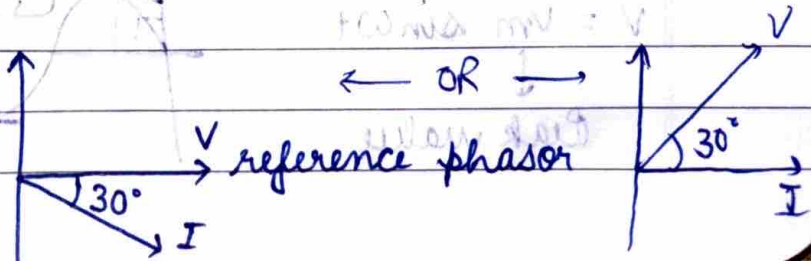


→ To trace several quantities in one graph, frequencies of all must be same

2. Phasor diagram

eg →

V leads current by 30°



• Mathematical representation

1. Rectangular form

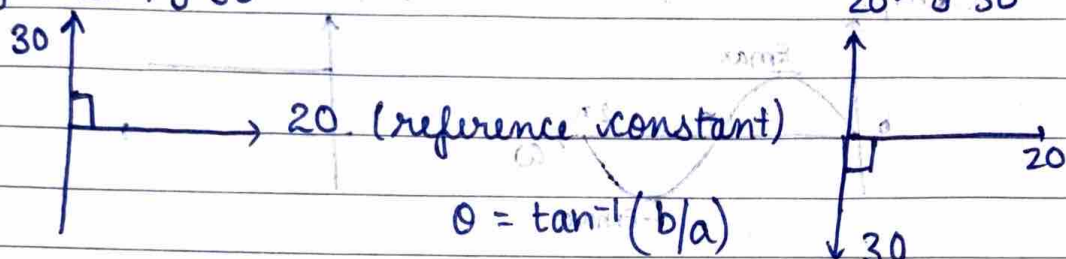
$$a \pm j b \quad (j = \sqrt{-1})$$

$a \rightarrow$ constant part $j \rightarrow$ phase shift of 90°

$b \rightarrow$ fluctuating part $\rightarrow$ lagging condition

$+$ $\rightarrow$ leading condition

eg $\rightarrow 20 + j 30$



2. Trigonometrical Form

$$a \cos \theta \pm j b \sin \theta$$

3. Polar Form

$$V = |V| \angle \pm \theta$$

eg $\rightarrow V = 20 \angle -30^\circ$

$|V| = 20V$ & it lags behind by 30°

• Magnitude Representation

1. Instantaneous

2. Peak / max value / amplitude

3. Average value

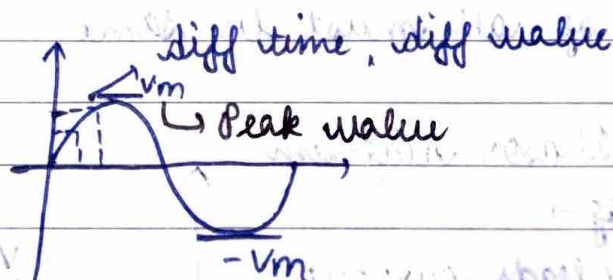
4. RMS value

1. Instantaneous

$$E = E_m \sin \omega t$$

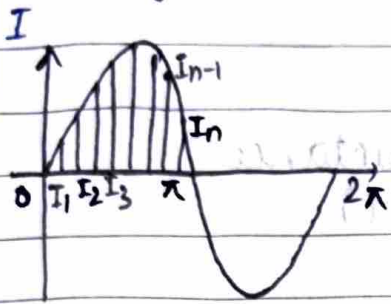
$$V = V_m \sin \omega t$$

$\downarrow$
Peak value





→ Average Value (Always consider half cycle)



$$\text{Avg} = \frac{\text{sum of all values}}{\text{no. of values}}$$

Taking area of alternation of $0-\pi$ and dividing it into all possible divisions

$$I_{\text{avg}} = \int_0^{\pi} \frac{I_m \sin \omega t}{\pi - 0} d(\omega t)$$

$$I_{\text{avg}} = \frac{I_m}{\pi} \int_0^{\pi} \sin \omega t d(\omega t)$$

$$= \frac{I_m}{\pi} (-\cos \omega t) \Big|_0^{\pi}$$

$$I_{\text{avg}} = \frac{2 I_m}{\pi}$$

~~~~~

→ RMS value (AC equivalent for D.C)

$$i = I_m \sin \omega t$$

SBS

$$i^2 = I_m^2 \left( \frac{1 - \cos 2\omega t}{2} \right)$$

Taking average of given value

$$\frac{I_m^2}{2} \int_0^{\pi} \frac{1 - \cos 2\omega t}{2} d(\omega t)$$

$$\frac{I_m^2}{2\pi} \left[ \omega t - \frac{\sin 2\omega t}{2} \right]_0^{\pi} = \frac{I_m^2}{2\pi} (\pi) = \frac{I_m^2}{2}$$

sq rooting both sides

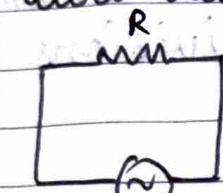
$$\Rightarrow I_{\text{rms}} = \frac{I_m}{\sqrt{2}}$$

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→ Analysis of single phase AC

Resistance (R) Capacitance (C) Inductance (L)

• Pure Resistive load

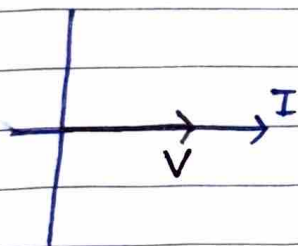


$$V = V_m \sin \omega t$$

Let's assume a purely resistive load across $V = V_m \sin \omega t$

$$i = \frac{V}{R} \quad \text{Ohm's law}$$

$$i = \frac{V_m \sin \omega t}{R} = I_m \sin \omega t$$



$$\text{Phase Angle} = 0$$

$$\text{Power dissipated} = V_r \times i$$

$$= V_m \sin \omega t \times I_m \sin \omega t$$

$$= V_m I_m \left(\frac{1 - \cos 2\omega t}{2} \right)$$

$$= \frac{V_m I_m}{2} - \frac{V_m I_m \cos 2\omega t}{2}$$

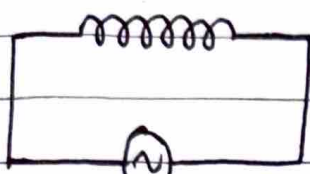
$$= 0 \quad \because \text{avg value} = 0$$

$$\Rightarrow \text{Power} = \frac{V_m I_m}{2} = \frac{V_m}{\sqrt{2}} \times \frac{I_m}{\sqrt{2}}$$

$$\text{Power} = I_{rms} \times V_{rms}$$



- Purely inductive load



$$V = V_m \sin \omega t$$

emf inside coil is $e = -N \frac{d\phi}{dt}$

Also $e = -V$ (Lenz law)

$$N = L$$

$$d\phi/dt = di/dt$$

convert into electrical units

$$\Rightarrow -V = -L \frac{di}{dt}$$

$$\Rightarrow V_m \sin \omega t = L \frac{di}{dt}$$

$$\Rightarrow \frac{V_m}{L} \sin \omega t \, dt = di$$

$$\Rightarrow i = \frac{V_m}{L} \int \sin \omega t \, dt$$

$$\Rightarrow i = -\frac{V_m}{\omega L} \cos \omega t$$

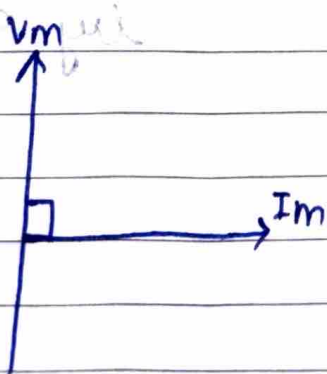
$$\Rightarrow i = \frac{V_m}{\omega L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

compare with Ohm's law

$$\omega L = X_L$$

$$\Rightarrow i = \frac{V_m}{X_L} \sin \left(\omega t - \frac{\pi}{2} \right)$$

$$\Rightarrow i = I_m \sin \left(\omega t - \frac{\pi}{2} \right) \rightarrow \text{Current lags voltage by } 90^\circ$$





$$\text{Power} = VI$$

$$= V_m \sin \omega t \cdot I_m \sin(\omega t - \pi/2)$$

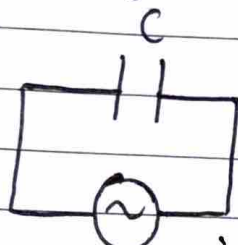
$$= -V_m I_m \sin \omega t \cos \omega t$$

$$= -\frac{V_m I_m}{2} \sin 2\omega t$$

$$\text{avg value} = 0$$

$$\Rightarrow \text{Power} = 0$$

• Purely Capacitive Load



$$V = V_m \sin \omega t$$

$$Q = CV$$

$$i = \frac{dQ}{dt}$$

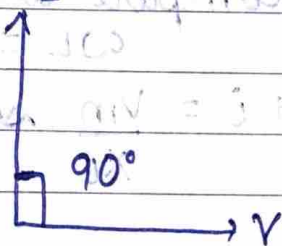
$$i = C \frac{dV}{dt}$$

$$i = C \frac{d(V_m \sin \omega t)}{dt}$$

$$i = C \omega V_m \cos \omega t = \omega C V_m \sin(\omega t + \pi/2)$$

$$i = \frac{V_m \sin(\omega t + \pi/2)}{1/\omega C}$$

$$\Rightarrow X_C = \frac{1}{\omega C}$$



$$\Rightarrow i = I_m \sin(\omega t + \pi/2) \rightarrow \text{Current leads by } 90^\circ$$

$$\text{Power} = V_m \sin \omega t \cdot I_m \cos \omega t$$

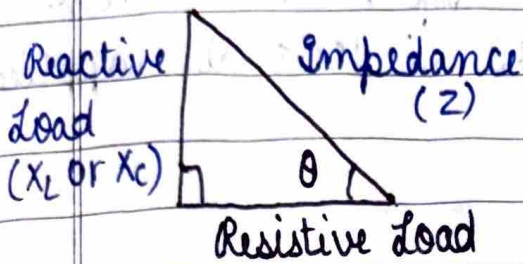
$$= \frac{V_m I_m}{2} \sin 2\omega t$$

$$\approx 0$$

$$\Rightarrow \text{Power} = 0$$



→ Impedance Triangle



$$Z = \sqrt{R^2 + (X_L - X_C)^2} \quad [X_L > X_C]$$

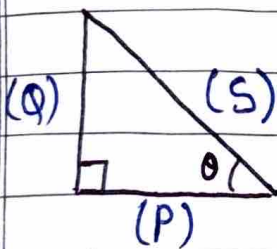
$$Z = \sqrt{R^2 + (X_C - X_L)^2} \quad [X_C > X_L]$$

$$Z = R \quad [X_L = X_C]$$

$$\sin \phi = \frac{|X_L - X_C|}{Z}$$

$$\cos \phi = \frac{R}{Z}$$

→ Power Triangle



$P \rightarrow$ Active Power $= V_{rms} I_{rms} \cos \phi$

$Q \rightarrow$ Reactive Power $= V_{rms} I_{rms} \sin \phi$

$S \rightarrow$ Apparent Power $= V_{rms} I_{rms}$

$P \rightarrow$ Watts

$Q \rightarrow$ volt-ampere-reactive

$S \rightarrow$ volt-ampere

→ Power Factor ($\cos \phi$)

Ratio of active & apparent power

$$\cos \phi = \frac{P}{S} = \frac{\text{active power}}{\text{apparent power}}$$



Q Find expression for current flowing & calculate power in series RL ckt of $R = 50 \Omega$ and $L = 0.159 \text{ H}$ when $V = 283 \sin 100t$ is applied

$$\omega = 100$$

$$X_L = \omega L = 15.9 \Omega$$

$$Z = \sqrt{R^2 + X_L^2} = \sqrt{2500 + 252.81} = 52.467 \Omega$$

$$I_0 = \frac{V_0}{Z} = \frac{283}{52.46} = 5.39 \text{ A}$$

$$I = 5.39 \sin(100t - \phi) \quad \because \text{current lags}$$

$$V_{\text{rms}} = \frac{283}{\sqrt{2}} = 200.11 \text{ V}$$

$$I_{\text{RMS}} = \frac{5.39}{\sqrt{2}} = 3.82 \text{ A}$$

$$\cos \phi = \frac{R}{Z} = \frac{50}{52.46}$$

$$\phi = 17.63^\circ$$

$$\Rightarrow I = 5.39 \sin(100t - 17.63)$$

NOW, average power & power factor

$$P = V_{\text{RMS}} \times I_{\text{RMS}} \times \cos \phi$$

$$= 200.11 \times 3.82 \times \frac{50}{52.46}$$

$$= 728.57 \text{ W}$$

Q Resistance of $15\ \Omega$ capacitor of $150\ \mu\text{F}$ connected in series across 230 V , 50 Hz supply

- a) Z b) I c) $\cos \phi$ d) ϕ e) P

$$R = 15\ \Omega$$

$$f = 50$$

$$\omega = 2\pi f = 100\pi$$

$$X_C = \frac{1}{\omega C} = \frac{1}{100\pi \times 150 \times 10^{-6}} = 21.23\ \Omega$$

$$a) Z = \sqrt{R^2 + X_C^2} = \sqrt{225 + 450.71} = 25.99\ \Omega$$

$$b) V_{\text{rms}} = 230\text{ V}$$

$$I_{\text{rms}} = \frac{230}{25.99} = 8.88\text{ A}$$

$$c) \cos \phi = \frac{R}{Z} = \frac{15}{25.99} = 0.577$$

$$d) \phi = \cos^{-1}(0.577) = 55.24^\circ$$

$$e) P = V_{\text{rms}} I_{\text{rms}} \cos \phi = 230 \times 8.88 \times 0.577 = 1178.46\text{ W}$$

Q 2 branches A & B are connected across supply of 200 V 50 Hz . If A consists in $R = 10\ \Omega$ and $L = 0.12\text{ H}$, B consists $R = 20\ \Omega$ & $C = 40\ \mu\text{F}$

a) Z

I_{total}

I in each branch

Power factor

v) Phasor diagram

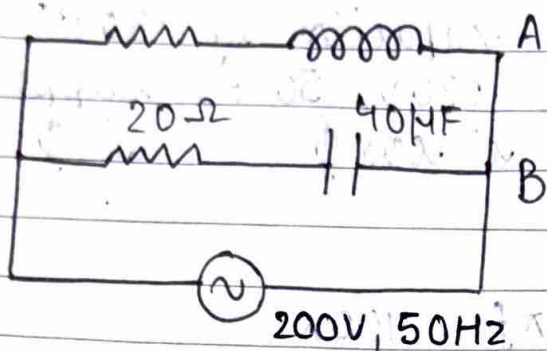
10- Ω

0.12H

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$$\omega = 2 \times 3.14 \times 50 = 314$$

$$X_L = \omega L = 314 \times 0.12 = 37.68 \Omega$$

$$X_C = 1/\omega C = 10^6/40 \times 314 = 79.61 \Omega$$

$$(R_{net})_A = \sqrt{100 + (37.68)^2} = 38.98 \Omega$$

$$(R_{net})_B = \sqrt{400 + (79.61)^2} = 82.08 \Omega$$

$$i) Z = \frac{R_A R_B}{R_A + R_B} = \frac{3199.47}{121.06} = 26.42 \Omega$$

$$ii) I_{net} = \frac{200}{26.42} = 7.57 A$$

$$iii) I_A = \frac{200}{38.98} = 5.13 A$$

$$I_B = \frac{200}{82.08} = 2.43 A$$

$$iv) \cos \phi = \frac{R}{Z} = \frac{R_{||}(10, 20)}{Z}$$

$$= \frac{20}{3 \times 26.42} = 0.252$$

$$\Rightarrow \phi = 75.404^\circ$$

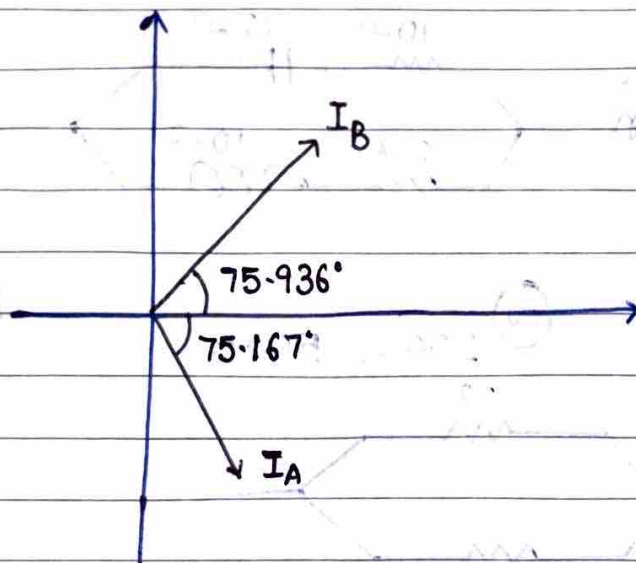


$$v) (\cos \phi)_A = \frac{10}{38.98}$$

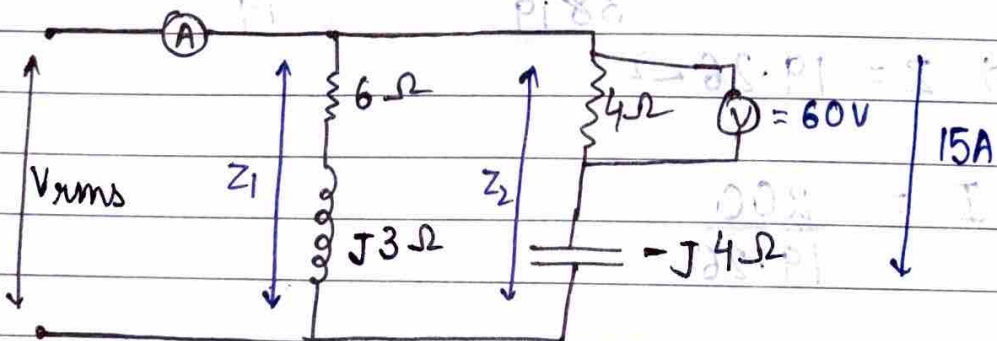
$$\Rightarrow \phi_A = 75.167^\circ$$

$$\text{||ly } (\cos \phi)_B = \frac{20}{82.08}$$

$$\Rightarrow \phi_B = 75.936^\circ$$



Q In given ckt, voltmeter reads 60V, find the ammeter reading



$$I_{4\Omega} = \frac{60}{4} = 15A$$

$$Z_1 = \sqrt{6^2 + 3^2} = \sqrt{36 + 9} = \sqrt{45} = 6.708 \Omega$$

$$Z_2 = 4 \angle 0^\circ = 5.656 \Omega$$

$$V_{rms} = 15 \times 5.656$$

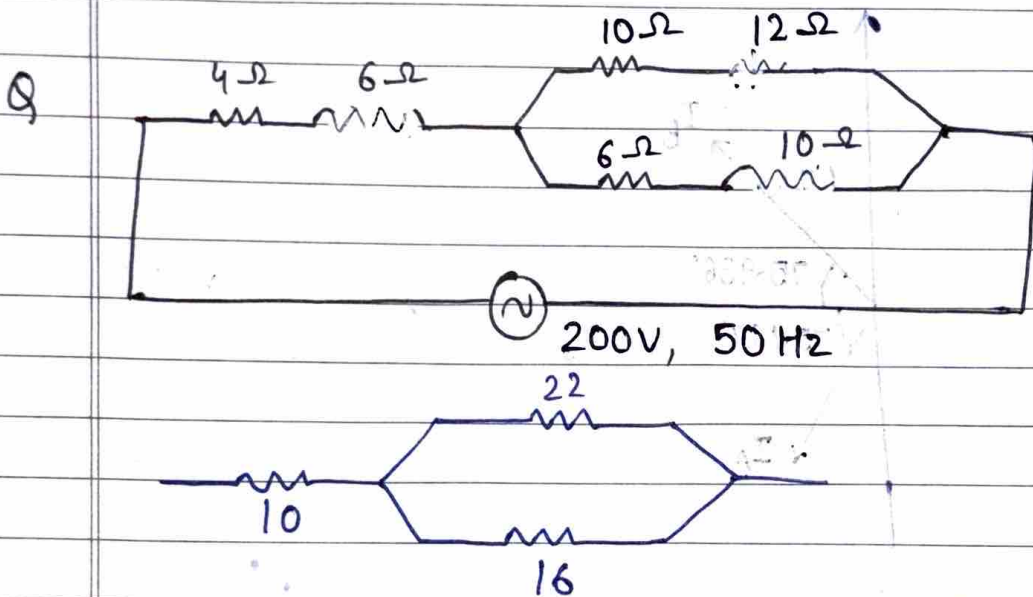
$$= 84.84 \text{ V}$$

$$\Rightarrow 84.84 = I_1 \times 6.708$$

$$\Rightarrow I_1 = 12.647 \text{ A}$$

$$I_{TOTAL} = 12.647 + 15 \text{ A}$$

$$= 27.647 \text{ A}$$



$$Z = 10 + \frac{22 \times 16}{38}$$

$$\frac{190 + 176}{19} = \frac{366}{19}$$

$$\Rightarrow Z = 19.26 \Omega$$

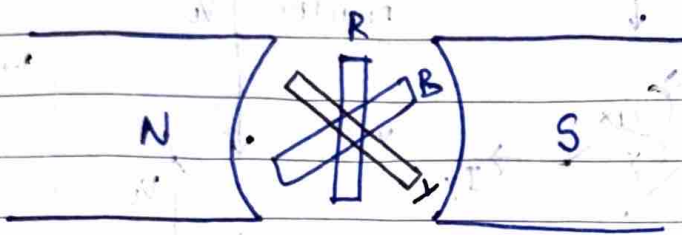
$$\Rightarrow I = \frac{200}{19.26}$$

$$= 10.38 \text{ A}$$



Three Phase A.C Circuits

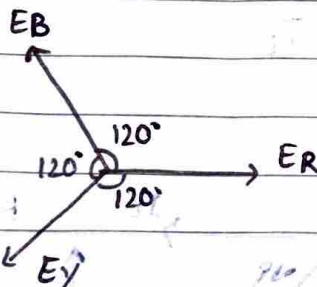
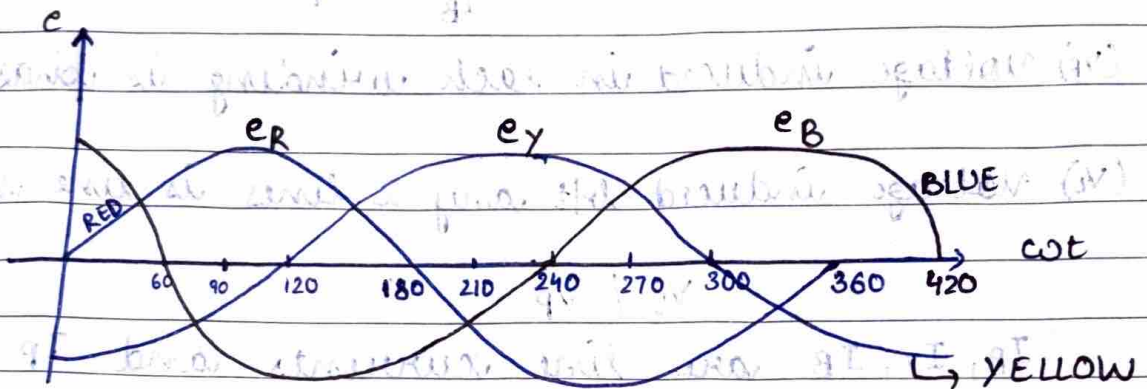
System producing three different alternating voltages



$$e_R = E_m \sin \omega t$$

$$e_B = E_m \sin (\omega t - 240^\circ)$$

$$e_Y = E_m \sin (\omega t - 120^\circ)$$

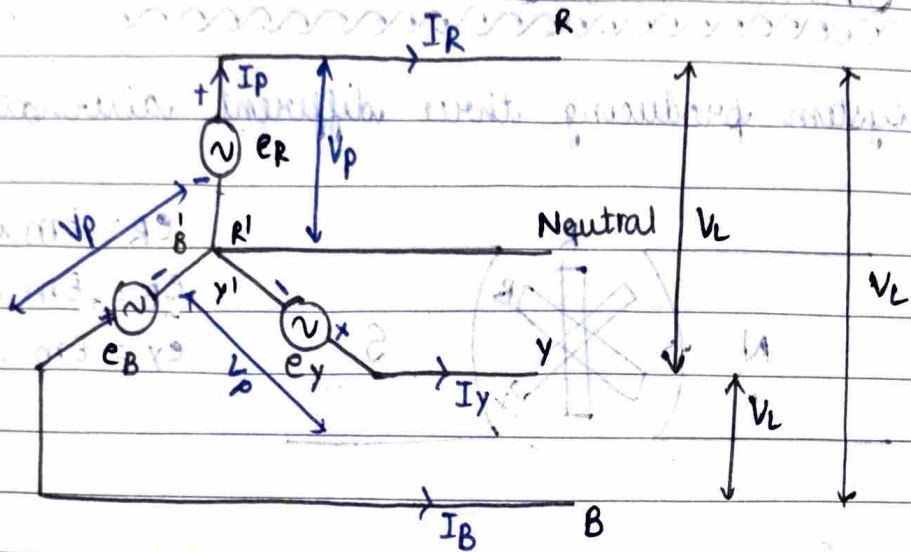


→ Benefits of 3 Phase

- constant power is delivered in 3 phase where it fluctuates in single phase
- more output as compared to single phase
- cheaper & economic power transmission
- more efficient
- high power factor
- small size



→ Star connection



(V_P) voltage induced in each winding is phase voltage

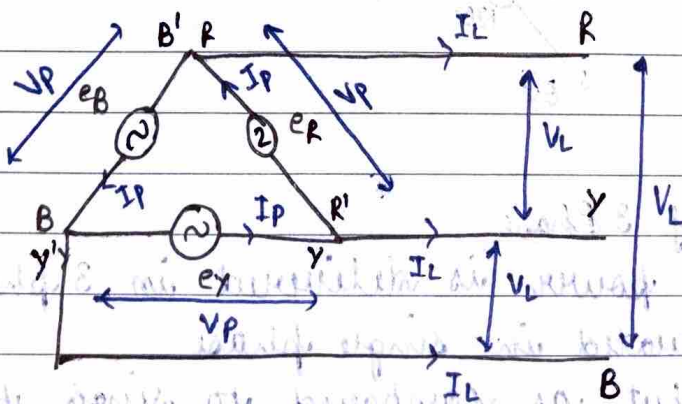
(V_L) voltage induced b/t any 2 lines is line voltage

$$V_L \neq V_P$$

I_R, I_Y, I_B are line currents and I_P is phase current

$$I_P = I_L$$

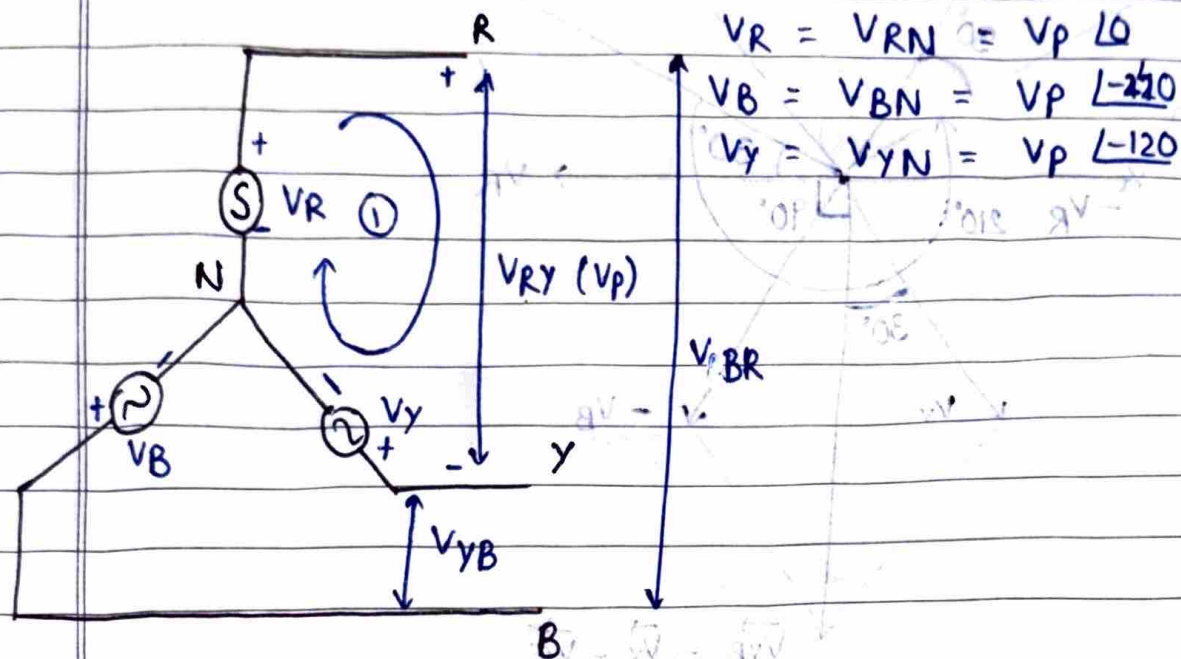
→ Delta connection



$$V_P = V_L$$

$$I_P \neq I_L$$

→ Relations of line & phase for star connection



KVL in ①

$$-\vec{V}_R + \vec{V}_{RY} + \vec{V}_Y = 0$$

$$\Rightarrow \vec{V}_{RY} = \vec{V}_R - \vec{V}_Y$$

$$= V_p \angle 0^\circ - V_p \angle -120^\circ$$

$$= V_p (\cos 0^\circ + j \sin 0^\circ) - V_p (\cos(-120^\circ) + j \sin(-120^\circ))$$

$$= V_p (1) - V_p (-1/2 - j\sqrt{3}/2)$$

$$= \frac{3V_p}{2} + \frac{j\sqrt{3}V_p}{2}$$

$$= \sqrt{3}V_p \left[\frac{\sqrt{3}}{2} + \frac{j}{2} \right] = \sqrt{3}V_p (\cos 30^\circ + j \sin 30^\circ)$$

$$= \sqrt{3}V_p \angle 30^\circ$$

$$\Rightarrow \underline{\underline{\vec{V}_{RY} = \sqrt{3} V_p \angle 30^\circ}}$$

$$\text{Similarly, } \vec{V}_{YB} = \vec{V}_Y - \vec{V}_B$$

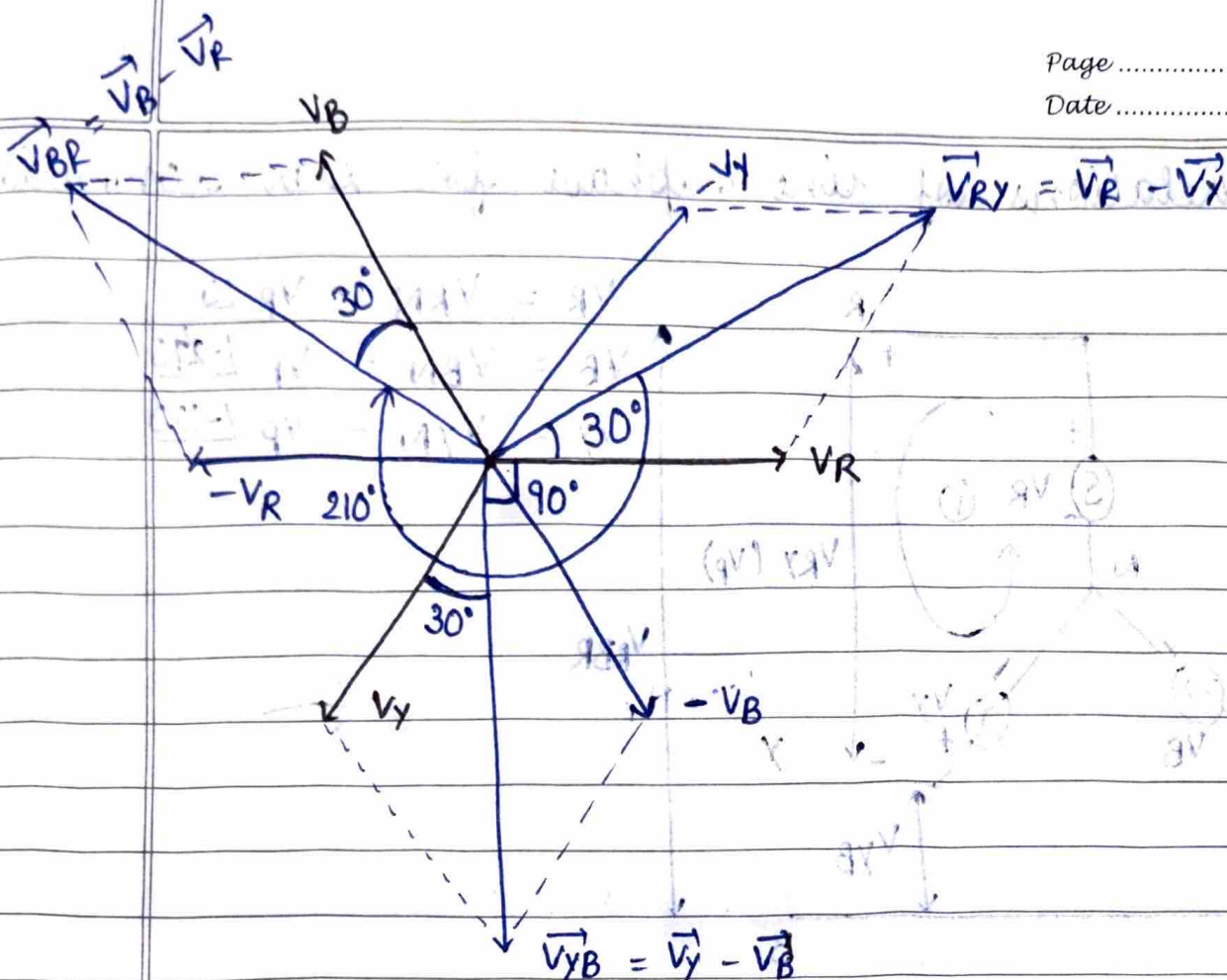
$$= V_p \angle -120^\circ - V_p \angle -240^\circ$$

$$\Rightarrow \underline{\underline{\vec{V}_{YB} = \sqrt{3} V_p \angle -90^\circ}}$$

$$\text{And } \vec{V}_{BR} = \vec{V}_B - \vec{V}_R$$

$$\Rightarrow \underline{\underline{\vec{V}_{BR} = \sqrt{3} V_p \angle 150^\circ}}$$

$$\Rightarrow \underline{\underline{\vec{V}_{BR} = \sqrt{3} V_p \angle -210^\circ}}$$



All line voltage leads respective phase voltage by 30° and magⁿ is $\sqrt{3}$ times

Hence,

$$\vec{V}_L = \sqrt{3} \vec{V}_P \angle 30^\circ$$

Also, $V_L = \sqrt{3} V_p$

$$(0.014 \text{ m} + 0.002 \text{ m}) \text{ gV} = \left[\frac{0.1}{2} + \frac{0.1}{2} \right] \text{ gV}$$

1970

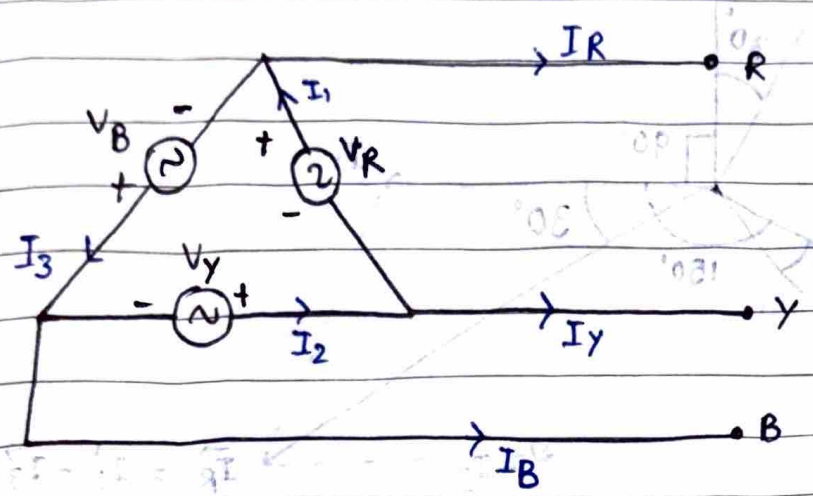
$$AKA = \sqrt{3} \sqrt{30}$$

$$gV - V = gV' + u''$$

0.05-1 qV - 0.05-1 qV -

$$\text{OP} \cdot \vec{qV} \cdot \vec{E} = qV \cdot \vec{E}$$

→ Relations for Delta connected



I_1, I_2, I_3 are phase currents
 I_R, I_Y, I_B are line currents

$$I_1 = I_p \angle 0^\circ$$

$$I_2 = I_p \angle -120^\circ$$

$$I_3 = I_p \angle -240^\circ$$

Using KCL

$$\vec{I}_1 = \vec{I}_3 + \vec{I}_R$$

$$\text{//ly } \vec{I}_Y = \vec{I}_2 - \vec{I}_1$$

$$\Rightarrow \vec{I}_R = \vec{I}_1 - \vec{I}_3$$

$$\vec{I}_B = \vec{I}_3 + \vec{I}_2$$

$$\vec{I}_R = I_p (\cos 0 + j \sin 0) - I_p (\cos -240 + j \sin 240)$$

$$\vec{I}_R = I_p (1) - I_p \left(-\frac{1}{2} + j \frac{\sqrt{3}}{2} \right)$$

$$\vec{I}_R = I_p \left(\frac{3}{2} - j \frac{\sqrt{3}}{2} \right) = \sqrt{3} I_p (\cos 30 - j \sin 30)$$

$$\vec{I}_R = \sqrt{3} I_p \angle -30^\circ$$

//ly, $\vec{I}_Y = \sqrt{3} I_p \angle -150^\circ$ and $\vec{I}_B = \sqrt{3} I_p \angle -270^\circ$



→ For star connection

$$V_L = \sqrt{3} V_P \quad I_L = I_P$$

$$\Rightarrow P = \sqrt{3} V_L I_L \cos \phi$$

$$Q = \sqrt{3} V_L I_L \sin \phi$$

$$S = \sqrt{3} V_L I_L$$

→ For delta connection

$$V_L = V_P \quad I_L = \sqrt{3} I_P$$

$$\Rightarrow P = \sqrt{3} V_L I_L \cos \phi$$

$$Q = \sqrt{3} V_L I_L \sin \phi$$

$$S = \sqrt{3} V_L I_L$$

→ Relationship between power in star & delta

Let's consider same load Z_P in both star and delta connections and V_L

Star connected

$$P = \sqrt{3} V_L I_L \cos \phi$$

$$\text{Now, } I_P = \frac{V_P}{Z_P} = \frac{V_L}{\sqrt{3} Z_P}$$

$$\Rightarrow I_L = I_P = \frac{V_L}{\sqrt{3} Z_P}$$

$$\Rightarrow P = \sqrt{3} V_L \times \frac{V_L}{\sqrt{3} Z_P} \cos \phi$$

$$\Rightarrow P_{\text{star}} = \frac{V_L^2}{Z_P} \cos \phi$$

Delta connected

$$I_P = \frac{V_P}{Z_P}$$

$$I_L = \sqrt{3} I_P$$

$$I_L = \sqrt{3} \frac{V_P}{Z_P} = \frac{\sqrt{3} V_L}{Z_P}$$

$$P = \sqrt{3} V_L \times \frac{\sqrt{3} V_L}{Z_P} \cos \phi$$

$$\Rightarrow P_{\text{delta}} = \frac{3 V_L^2}{Z_P} \cos \phi$$

$$\underline{P_{\text{delta}} = 3 \times P_{\text{star}}}$$



→ Balanced star to delta or delta-star conversions

$V_L I_L \rightarrow$ line quantities

$Z_Y \rightarrow$ phase impedance of star

$Z_\Delta \rightarrow$ phase impedance of delta

$$Z_Y = \frac{V_P}{I_P} = \left(\frac{V_L}{\sqrt{3}} \right) \times \frac{1}{I_L}$$

$$Z_Y = \frac{1}{\sqrt{3}} \frac{V_L}{I_L}$$

$$Z_\Delta = \frac{V_P}{I_P} = \frac{V_L(\sqrt{3})}{I_L} = \sqrt{3} \frac{V_L}{I_L}$$

Star connection

$$V_L = \sqrt{3} V_P$$

Delta connection

$$V_L = V_P$$

1.

$$I_L = I_P$$

$$I_L = \sqrt{3} I_P$$

2. V_L leads V_P by 30°

V_L lags V_P by 30°

$$P_Y = P_\Delta / 3$$

$$P_\Delta = 3 P_Y$$

3. 3 phase 3 wire or

3 phase - 3 wire

3 phase 4 wire

system

system

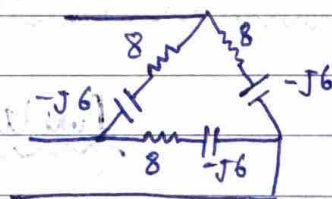
Q A balanced delta connected load of impedances $(8 - j6) \Omega$ per phase is connected to a 3 ϕ 230V, 50 Hz supply

a) Power factor

b) Line current

c) Reactive power

$$Z_P = \sqrt{8^2 + 6^2} = 10 \Omega$$

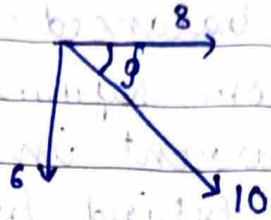




$$V_L = 230V$$

$$I_L = \sqrt{3} I_P \angle -30^\circ$$

$$I_P = \frac{V_P}{Z_P} = \frac{230}{8 - j6}$$



$$\cos \phi = 0.8$$

$$\phi = 36.87^\circ$$

$$\Rightarrow \vec{I}_P = 23 \angle 36.87^\circ \text{ A} \quad \text{or} \quad I_P = 23$$

$$\Rightarrow \text{Power factor} = \cos \phi = 0.8$$

$$\Rightarrow \vec{I}_L = \sqrt{3} \times 23 \angle 36.87^\circ - 30^\circ$$

$$= \sqrt{3} \times 23 \angle 6.87^\circ \text{ A}$$

$$\text{or } I_L = 23\sqrt{3} = 39.84$$

$$\Rightarrow Q = \sqrt{3} V_L I_L \sin \phi$$

$$= \sqrt{3} \times 230 \times 39.84 \times 0.6$$

$$= 9.52 \text{ KVar}$$

Q 3 equal impedences each of $10 \angle 60^\circ \Omega$ are connected in star across 3ϕ , $400V$, 50Hz supply. Calculate V_P & V_L , power factor, active power. If same impedance are connected in delta to same source what is active power?

$$V_L = 400V$$

$$V_P = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94V$$

$$I_P = \frac{V_P}{Z_P} = \frac{230.94}{10 \angle 60^\circ} = 23.094 \angle -60^\circ \text{ A}$$

$$\Rightarrow I_P = 23.094 \angle -60^\circ \text{ A} = I_L$$

$$P = \sqrt{3} \times 400 \times 23.094 \times \cos 60^\circ = 7.999 \text{ kW} \approx 8 \text{ kW}$$

$$P_{\Delta} = 3 P_{\gamma} = 3 \times 8 = 24 \text{ kW}$$

Q A balanced star connected load is supplied from symmetrical 3ϕ , $400V$, $50Hz$ systems. Current in each phase is $30A$ and lags 30° behind by V_p

- V_p
- Resistance & Reactance per phase
- Load inductance per phase
- Total power consumed

$$V_L = 400 V \quad f = 50Hz$$

$$a) V_p = \frac{V_L}{\sqrt{3}} = \frac{400}{\sqrt{3}} = 230.94 V$$

$$b) Z_p = \frac{230.94}{30 \angle -30^\circ} = 7.698 \angle 30^\circ \Omega$$

$$\Rightarrow Z_p = 7.698 \left(\frac{\sqrt{3}}{2} + j \frac{1}{2} \right) \Omega$$

$$\Rightarrow Z_p = 6.67 + j 3.849 \Omega$$

$\Rightarrow$ Resistance per phase Reactance (X_p) per phase

$$c) X_p = 2\pi f L$$

$$L = \frac{X_p}{2\pi f} = \frac{3.849 \times 7}{100 \times 22} = 0.0122 H$$

$$d) P = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} \times 400 \times 30 \times \cos(-30^\circ)$$

$$= 18.18 KW$$



- Q A symmetrical 3 ϕ 400V system supplies a load at 0.8 lagging power factor which is connected in star. Current drawn by the impedance is 34.64 A.
- impedance
 - resistance & reactance per phase
 - total active & reactive power

$$V_L = 400 \text{ V}$$

$$I_L = 34.64$$

$$\cos \phi = 0.8$$

$$\vec{I}_p = \vec{I}_L \angle -36.87^\circ$$

$$\Rightarrow \phi = 36.87^\circ$$

$$\Rightarrow Z_p = \frac{V_p}{I_p} = \frac{400}{\sqrt{3} \times 34.64} \angle -36.87^\circ$$

$$\Rightarrow Z_p = 6.67 \angle 36.87^\circ \Omega$$

$$= 6.67 (0.8 + j0.6) \Omega$$

$$\Rightarrow Z_p = 5.33 + j4.005 \Omega$$

$$\Rightarrow R_p = 5.33 \Omega$$

$$\Rightarrow X_p = 4.005 \Omega$$

$$c) P = \sqrt{3} V_L I_L \cos \phi$$

$$= \sqrt{3} \times 400 \times 34.64 \times 0.8$$

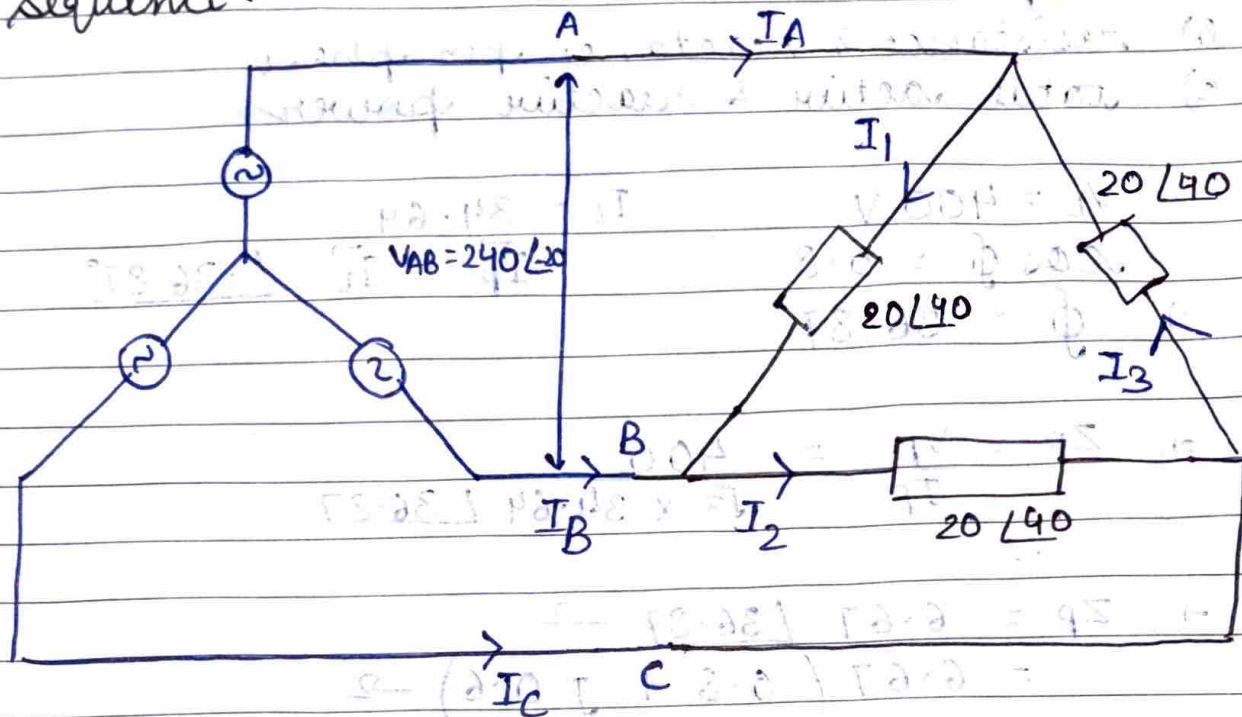
$$= 19.199 \text{ kW}$$

$$Q = \sqrt{3} V_L I_L \sin \phi$$

$$= \sqrt{3} \times 400 \times 34.64 \times 0.6$$

$$= 14.399 \text{ KVar}$$

Q. The line voltage of a balanced star connected load is $V_{AB} = 240 \angle -20^\circ \text{ V}$. If source is connected to Δ connected load of $20 \angle 40^\circ \Omega$. Find phase & line currents. Assume abc sequence.



$$I_1 = \left(\frac{V_p}{Z_p / \Delta} \right) = \frac{240 \angle -20^\circ}{20 \angle 40^\circ} = 12 \angle -60^\circ$$

$$I_2 = I_1 \angle -120^\circ = 12 \angle -180^\circ$$

$$I_3 = I_1 \angle -240^\circ = 12 \angle -300^\circ$$

Now, $I_A = \sqrt{3} I_1 \angle -30^\circ$ (line lags in Δ)

$$I_A = \sqrt{3} \cdot 12 \angle -60^\circ \angle -30^\circ$$

$$I_A = 12\sqrt{3} \angle -90^\circ$$

$$I_A = 20.79 \angle -90^\circ \text{ A}$$

$$\Rightarrow I_B = I_A \angle -120^\circ = 20.79 \angle -210^\circ$$

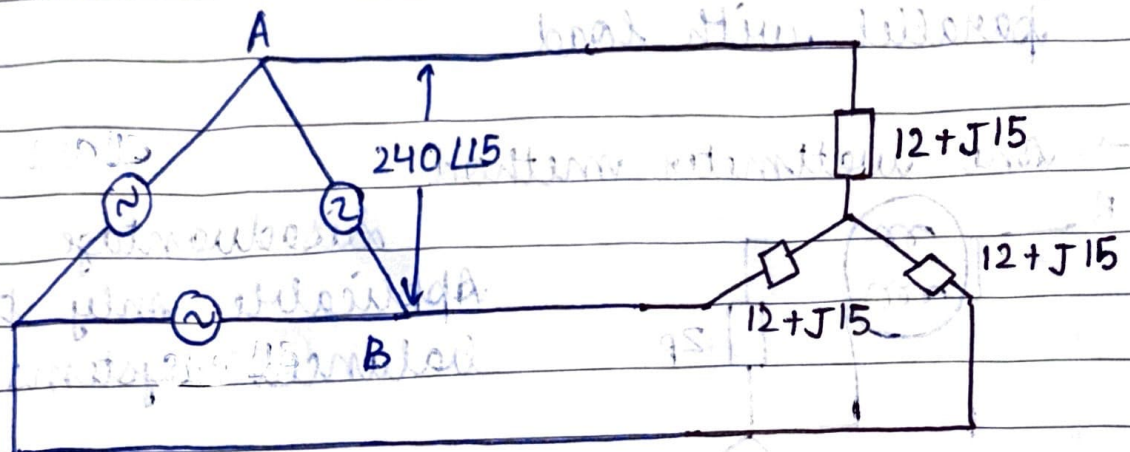
$$\Rightarrow I_C = I_A \angle -240^\circ = 20.79 \angle -330^\circ$$



Q 3n balanced Δ - γ circuits $V_{AB} = 240 \angle 15^\circ \text{ V}$
 $Z_\gamma = (12 + j15) \Omega$ calculate line currents

supply $\rightarrow \Delta$

load $\rightarrow \gamma$



$$V_L = 240 \angle 15^\circ$$

$$\text{3n star} = V_L = \sqrt{3} V_p \angle 30^\circ$$

$$\Rightarrow V_p = \frac{V_L}{\sqrt{3}} \angle -30^\circ$$

$$\Rightarrow V_p = \frac{240}{\sqrt{3}} \angle -15^\circ = 138.56 \angle -15^\circ$$

$$I_A = \frac{(V_p)_A}{(Z_p)_\gamma} = \frac{138.56 \angle -15^\circ}{12 + j15} = 7.213 \angle -66.34^\circ \text{ A}$$

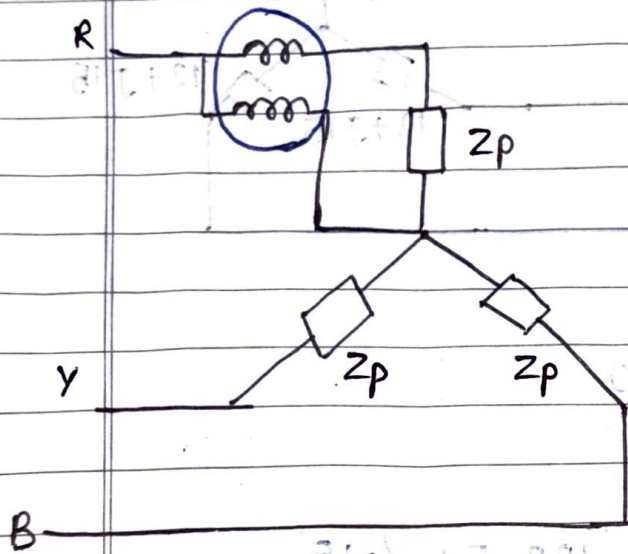
$$I_B = I_A \angle -120^\circ = 7.213 \angle -186.34^\circ \text{ A}$$

$$I_C = I_A \angle -240^\circ = 7.213 \angle -306.34^\circ \text{ A}$$



- Measurement of 3 ϕ power
- Wattmeter is used, it has two coils
1. current coil which is connected in series with load
 2. potential coil which is connected in parallel with load

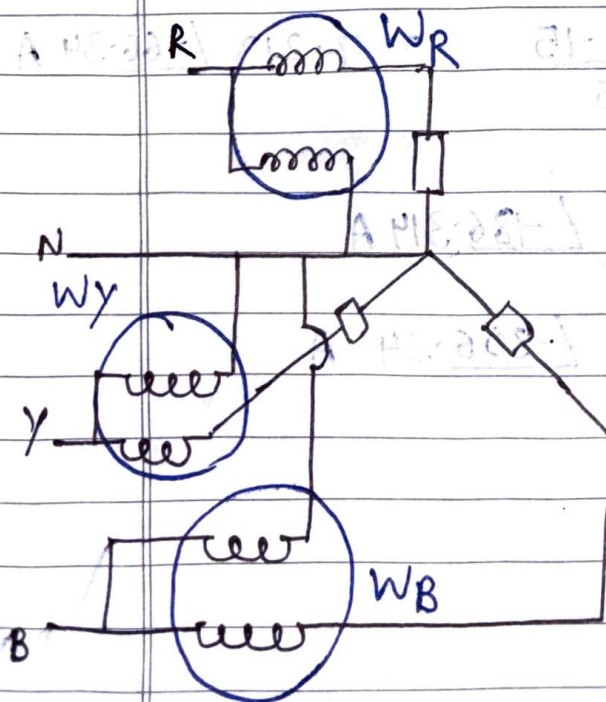
→ One wattmeter method



Disadvantage
Applicable only to balanced systems

Advantage
It is economical

→ Three wattmeter method

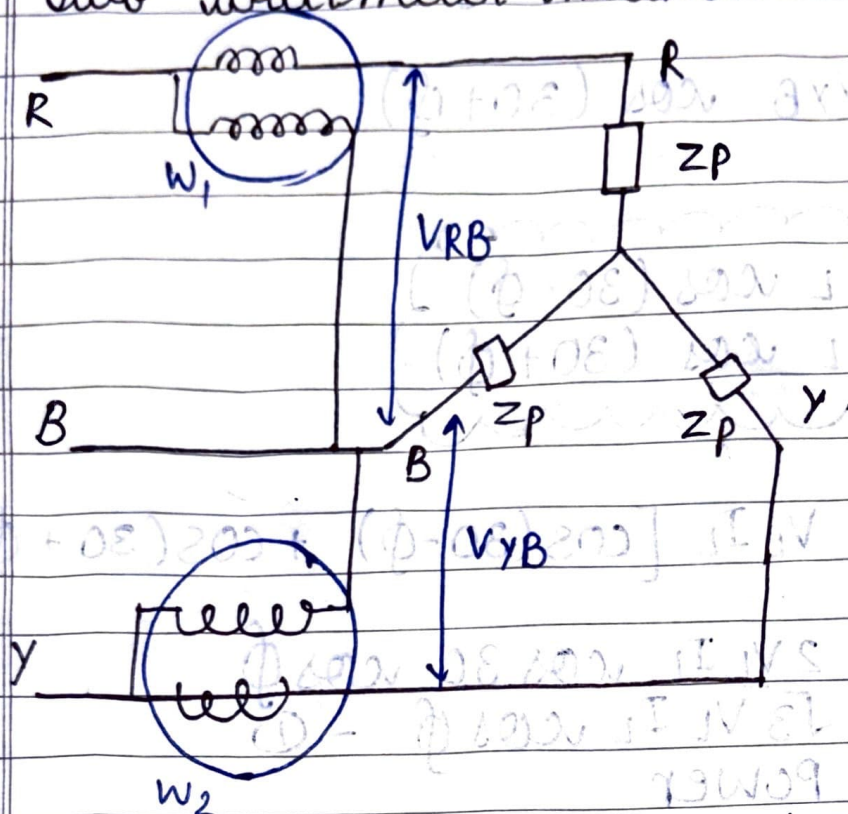


$$P_{net} = W_1 + W_2 + W_3$$

Applicable to both balanced & unbalanced but is very costly



→ Two wattmeter method

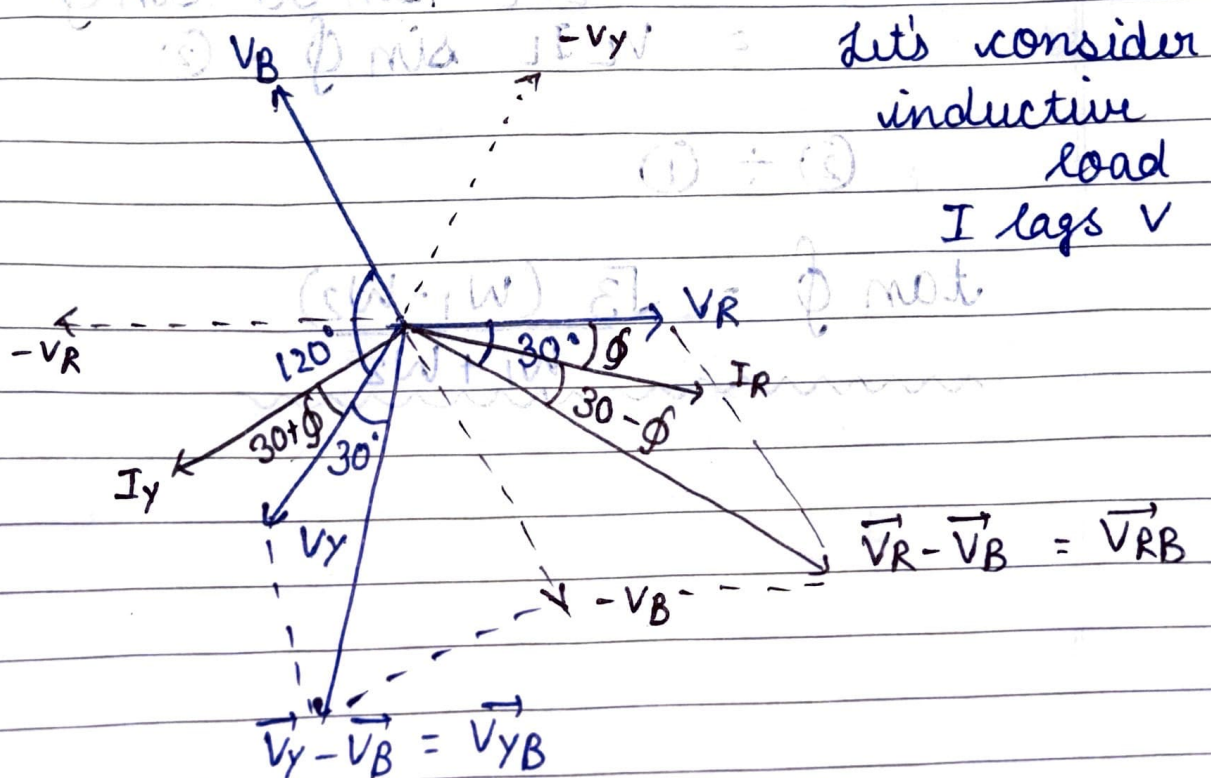


$$W_1 = I_R \times V_{RB} \cos(\text{angle b/t } I_R \text{ \& } V_{RB})$$

$$W_2 = I_Y \times V_{YB} \cos(\text{angle b/t } I_Y \text{ \& } V_{YB})$$

(30 - ϕ)

(30 + ϕ)



$$\Rightarrow W_1 = I_R V_{RB} \cos(30^\circ - \phi)$$

$$\Rightarrow W_2 = I_Y V_{YB} \cos(30^\circ + \phi)$$

$$\Rightarrow W_1 = V_L I_L \cos(30^\circ - \phi)$$

$$W_2 = V_L I_L \cos(30^\circ + \phi)$$

$$W_1 + W_2 = V_L I_L [\cos(30^\circ - \phi) + \cos(30^\circ + \phi)]$$

$$= 2 V_L I_L \cos 30^\circ \cos \phi$$

$$= \sqrt{3} V_L I_L \cos \phi \quad \text{--- (1)}$$

$$= \text{Power}$$

$$\Rightarrow \text{Active Power} = W_1 + W_2$$

$$\Rightarrow W_1 - W_2 = 2 V_L I_L \sin 30^\circ \sin \phi$$

$$= V_L I_L \sin \phi \quad \text{--- (2)}$$

$$(2) \div (1)$$

$$\tan \phi = \frac{\sqrt{3} (W_1 - W_2)}{W_1 + W_2}$$